Analytical approach of the effect of surfactant on Mixed Metal Oxide Materials (ZnO/NiO) to enhance the sensitivity of a humidity sensor

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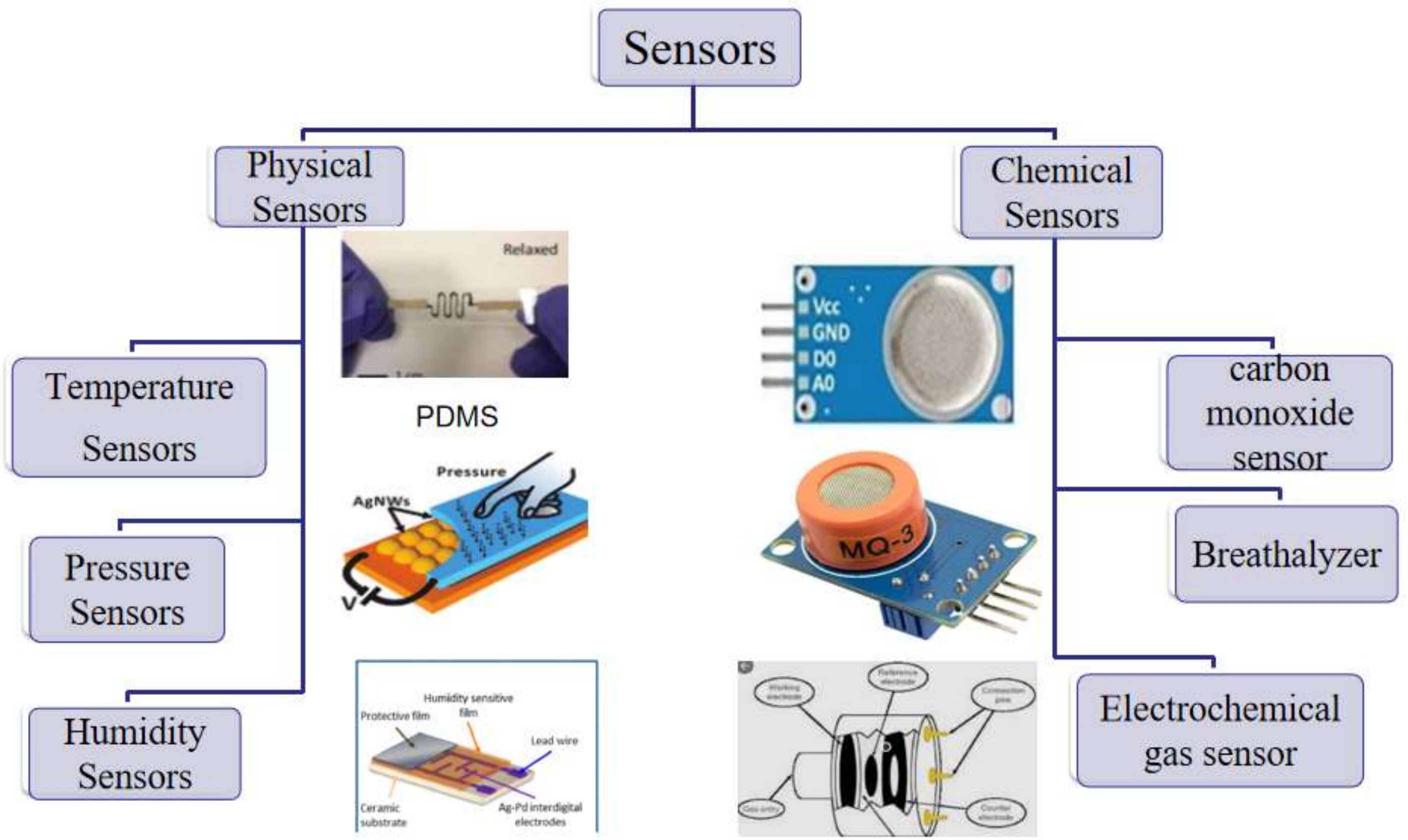
Introduction

- **Sensors** : Response to the stimulus (heat, light, sound, pressure) in the form of electrical signals

- **Classifications:**



Humidity Sensor

- **Definition:**

  A humidity sensor is an electronic device that measures the humidity in its environment and converts its findings into a corresponding electrical signal

- **Types:** 1. Relative humidity (RH) sensors and

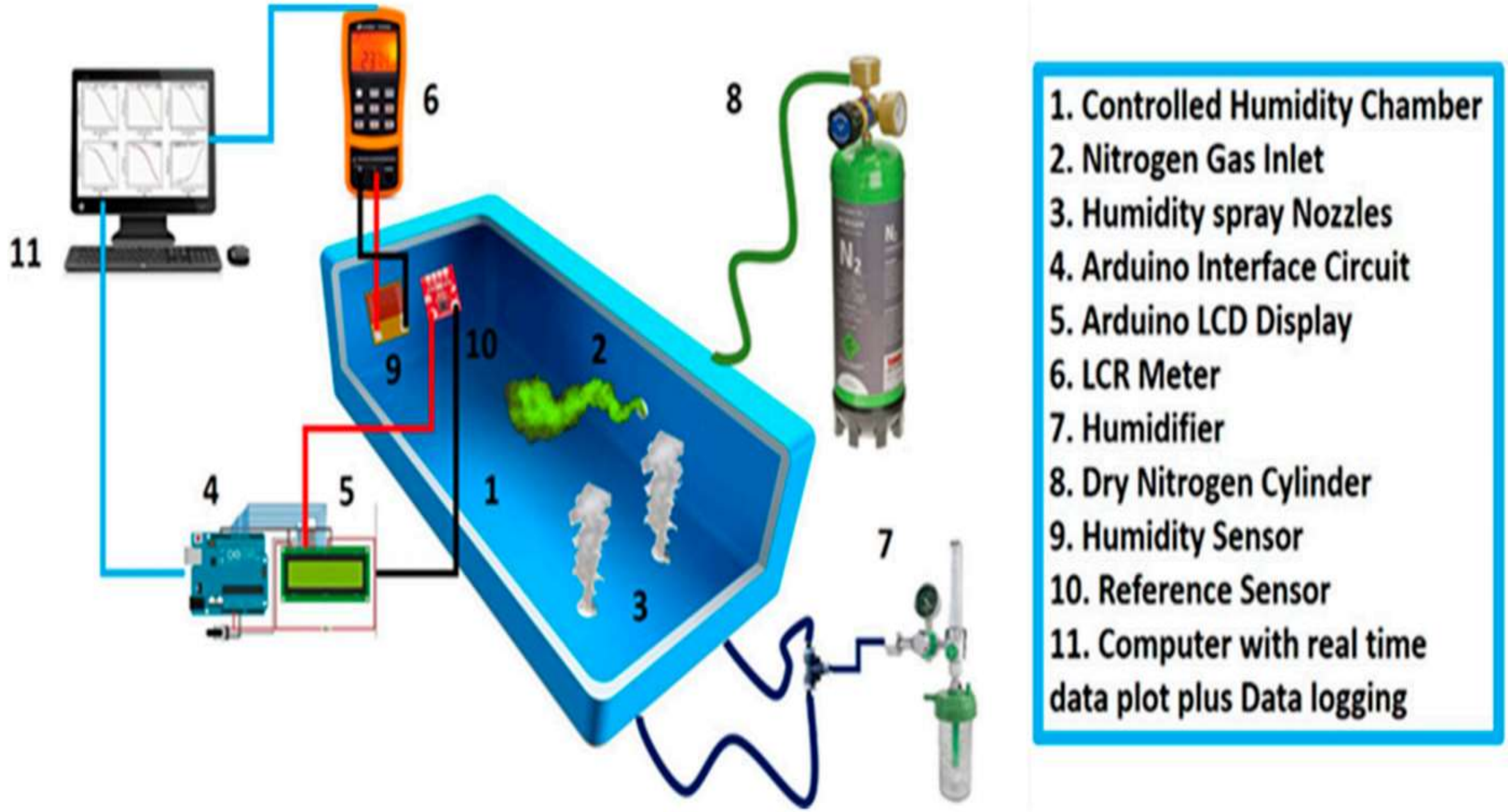
  2. Absolute humidity (AH) sensors

Relative humidity (%)	Absolute humidity
The amount of water vapour present in air expressed as a percentage of the amount needed for saturation at the same temperature	The amount of water vapour in 1cu.m. of air

- **Applications:** Humidity sensors are commonly used in

  Meteorology
  Medical
  Automobile
  HVAC and
  Manufacturing industries

Working Principle



R.H Sensing Mechanism: Nanomaterials are chosen in such a way that the resistivity varies with respect to the variation of relative humidity

Nanostructure

- Describes the molecular level of materials typically between 1nm to 100nm(1nm=10^{-9}m) in size.
- Advantages:
  1. Smaller
  2. Faster
  3. Lighter
  4. Cheaper
  5. Less Raw material
  6. Consumes very less energy and
  7. Surface to Volume ratio

Metal Oxide

Metal oxides are crystalline solids that contain a metal cation and an oxide anion

The properties are:

* Stability - Due to its electro - negativity
* High reactivity - By electrolytic reduction
* High Selectivity - Ratio of the desired product formed to the undesired product formed
* Regenerability - The action or process of regenerating polymer fibers
* Robustness
* Low cost

Why Oxides?

Oxides provide the below mentioned crucial properties:

- Strong interplay b/w spin/charge/lattice
- They have a wide variety of crystal structure
- They can have the “Nano structuration”
- They have “ Transport properties”
- They are used in “visible Spectrum”
- They can be doped for altering properties like

  i) Optical

  ii) Structural

  iii) Morphological

  iv) Electrical

Choice of Metal Oxide

Materials

MgO | CuO | Fe_2O_3 | ZnO | SnO | Al_2O_3 | CO_3O_4







- Magnesium oxide (MgO)
- Copper oxide (CuO)
- Ferric Oxide (Fe_2O_3)
- Zinc oxide (ZnO)
- Tin-oxide (SnO)
- Aluminium Oxide (Al_2O_3)
- Cobalt Oxide (CO_3O_4) etc…

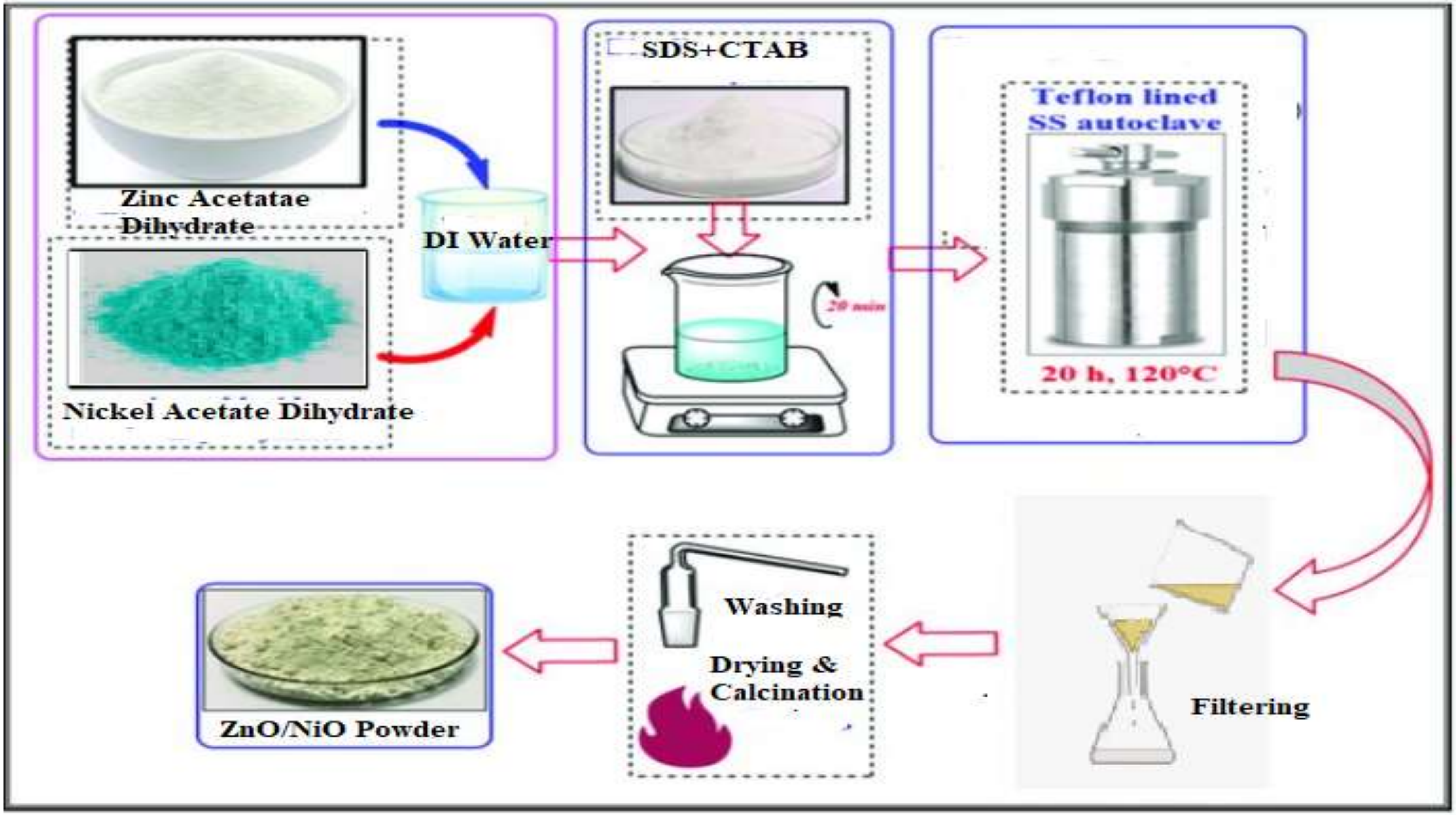
Why ZnO?

- Good transparency
- High electron mobility
- Wide band gap (3.4 eV, Direct)
- Strong room-temperature luminescence
- Boiling point: 2360 °C
- Melting point: 1975 °C
- Large exciton binding energy of 60 meV
- Nontoxic
- Not react with food or container

Methods of Fabrication

- There are 3 methods used:
    1. Hydrothermal Method
    2. Doctor blade method
    3. Spin Coating

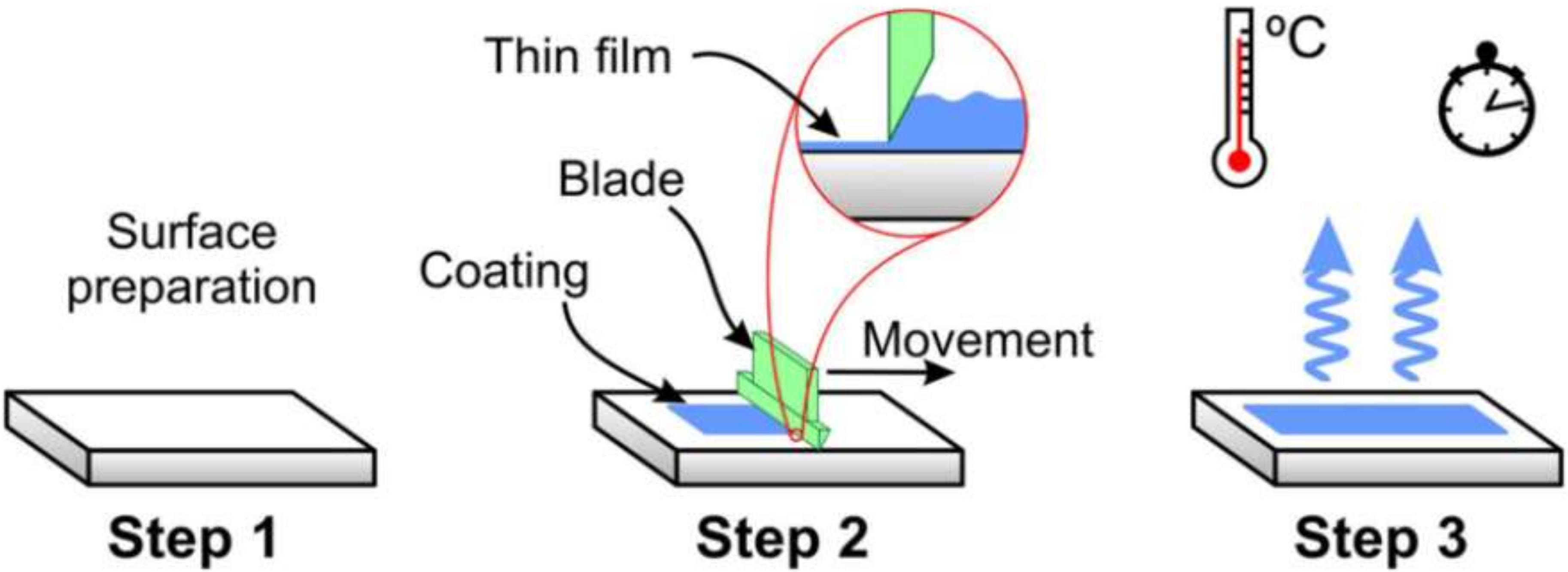
1. Hydrothermal Method:



- It refers to the synthesis of substances via chemical reactions in a sealed and heated solution above ambient temperature and pressure

Fig: Illustration of Hydrothermal method

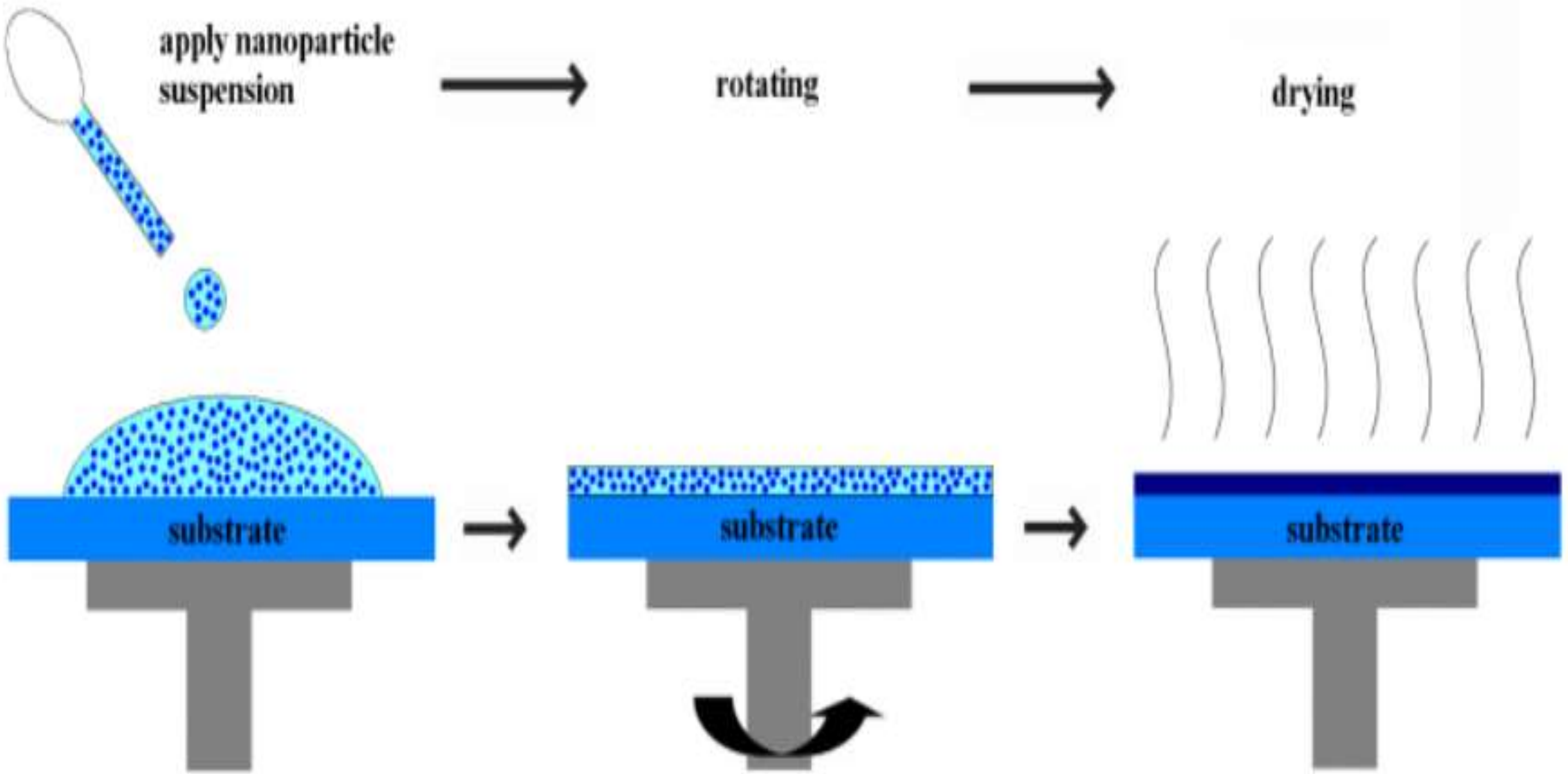
2. Doctor blade method:



- Doctor blade coating is a technique used to form films with well-defined thicknesses. This technique works by placing a sharp blade at a fixed distance from the surface that needs to be covered.

Fig: Illustration of Doctor blade method

3. Spin Coating:



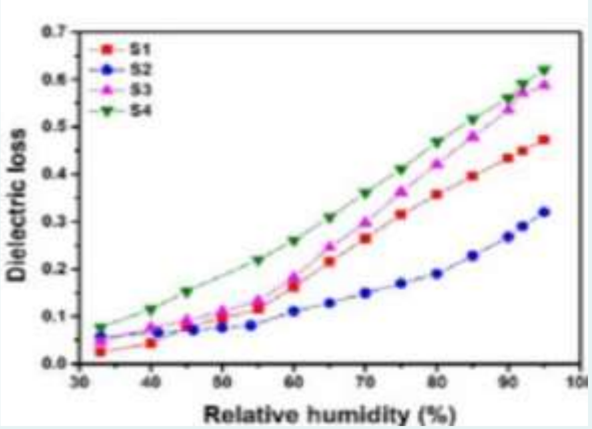
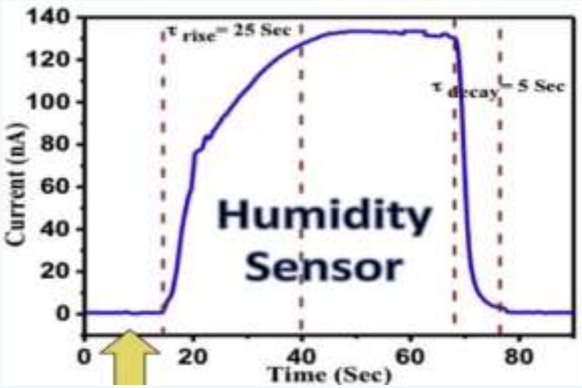
- It is a procedure used to deposit uniform thin films onto flat substrates.

Fig: Illustration of Spin Coating method

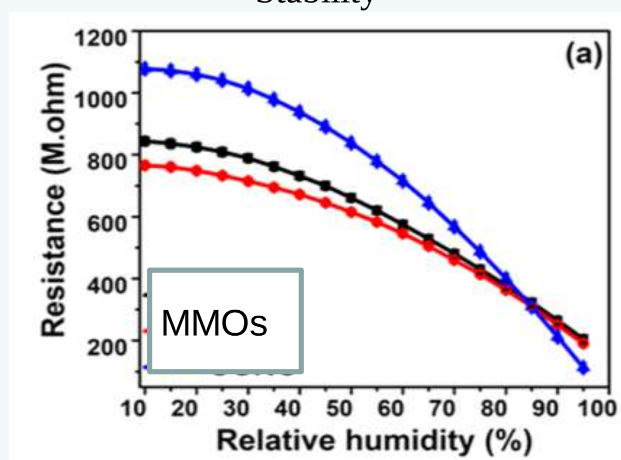
Research Objectives

- To fabricate a surfactant based humidity sensor with better sensitivity
- To increase the sensitivity of a humidity sensor by adding Hexadecyltrimethylammonium bromide and Sodium Dodecyl Sulphate (CTAB: SDS)
- To verify the performance of a humidity sensor through a resistive change in accordance with the instantaneous change in relative humidity

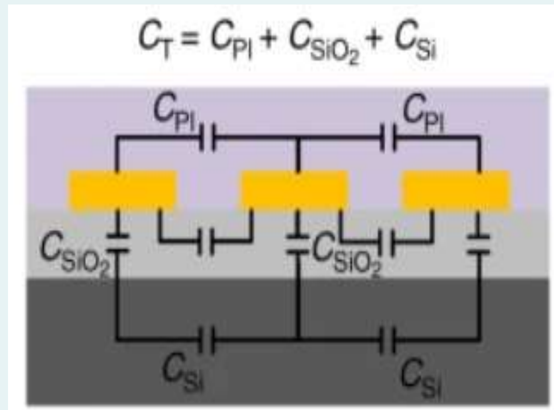
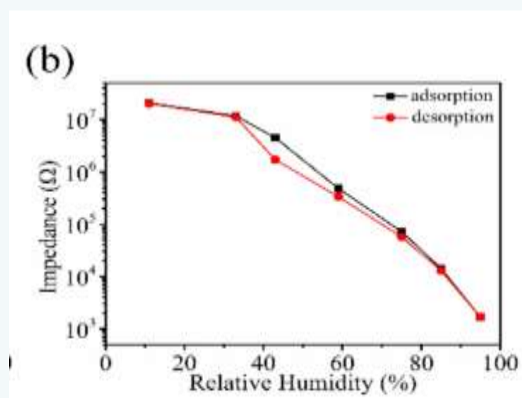
Literature Survey

S.NO	PUBLICATION DETAILS	MATERIAL/PARAMETER	INFERENCE
1	Anamitra Chattopadhyay "Improvement of humidity sensing performance ad dielectric response through pH variatios in ceramics" 2022	Calcium copper titanate /Relative Humidity and Pore size 	• Exhibited maximum sensitivity in the relative humidity (RH) range 33–95%. • Various factors such as porosity, and grain sizes contributing to the sesnor response.
2	Shobhnath P. Gupta "Superior humidity sensor and photodetector of mesoporous ZnO nano sheets at room temperature" 2019	ZiO NanoSheets/Response and Recovery Time 	• Thickness of Nano sheets is around 18–22 nm. • Further the mesoporous ZnO NS with the pore size between 5–10 nm • Response time: 25s • Recovery time: 5 s

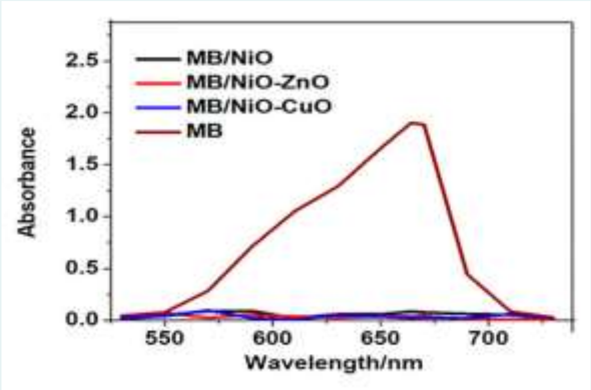
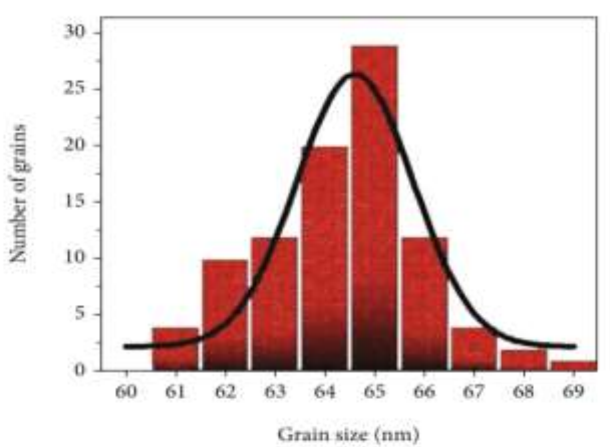
Literature Survey

S.NO	PUBLICATION DETAILS	MATERIAL/PARAMETER	INFERENCE
3	Sonima Mohan "Hydrothermal synthesis and characterization of Zinc Oxide nanoparticles of various shapes under different reaction conditions"2021	ZnO/Size and shape of nanomaterials 	• Morphology depends on reaction time, temperature and Concentration of reacting solutions
4	Kausar Shaheen "Electrical, Photocatalytic, and Humidity Sensing Applications of Mixed Metal Oxide Nanocomposites" 2020	MMO/Response, Recovery Time and Stability 	• Electrical and electronic devices are operated on the basis of dielectric characteristics • The dielectric constant decreased with the increase in the frequency • There is a change in resistance w.r.t the variations in humidity levels $S_H = \frac{\Delta R_o}{\Delta R_{RH}}$

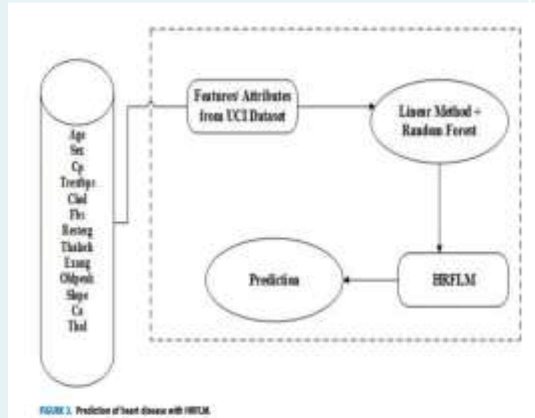
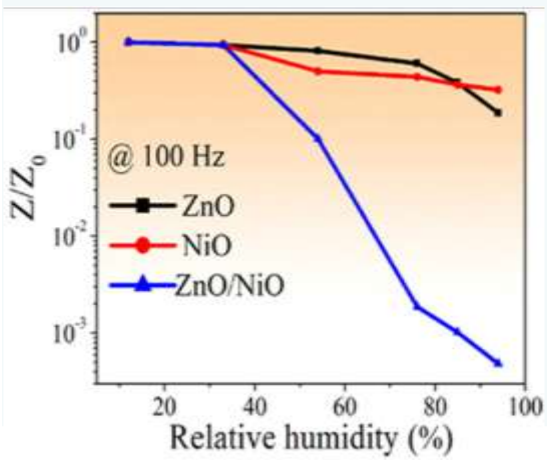
Literature Survey

S.NO	PUBLICATION DETAILS	MATERIAL/PARAMETER	INFERENCE
5	Guidong Chen "A nanoforest-based humidity sensor for respiration monitoring" 2022	3-Layer Capacitance/Paracitic Capacitance, Temperature drift  	• Existence of these parasitic capacitances results in the inaccurate determination of the humidity-sensitive layer capacitance • Recovery time to 5 s. • Machine learning algorithms- sensor can distinguish different respiration states with an accuracy of 94%.
6	Fan Li "Preparation and Research of a High-Performance ZnO/SnO2 Humidity Sensor" 2022	(ZnO/SiO_2)/Resistance sensor hysteresis loop  	• Response and recovery speeds (35 and 8 s, respectively • Response = $R = \frac{R_L - R_H}{R_H} \times 100\%$ • Hysteresis Error= $\delta H = \frac{\Delta H_{max}}{2F_{FS}}$

Literature Survey

S.NO	PUBLICATION DETAILS	MATERIAL/PARAMETER	INFERENCE
7	Hulugirgesh Degefu Weldekirstos "Surfactant-Assisted Synthesis of NiO-ZnO and NiO-CuO Nanocomposites for Enhanced Photocatalytic Degradation of Methylene Blue Under UV Light Irradiation"2022	ZnO/Surface area 	• Surface properties - morphological structure, specific surface area, and particle size • Energy band gaps of pure NiO, NiO-ZnO, and NiO-CuO composites are : 3.0, 2.9, and 3.25 eV, respectively.
8	Jingyu Wang "Research Progress on Humidity-Sensing Properties of Cu-Based Humidity Sensors" 2022	CuO/Surface modification and nanocomposites 	• Sensing mechanism:Change in the resistance of the sensing element. • Physicochemical reaction causes the impedance change. • grain size affecting its moisture-sensing properties. • Particle size is reduced:specific surface area improved • Material synthesis: affects surface morphology, increasing the surface porosity and increasing the grain size.

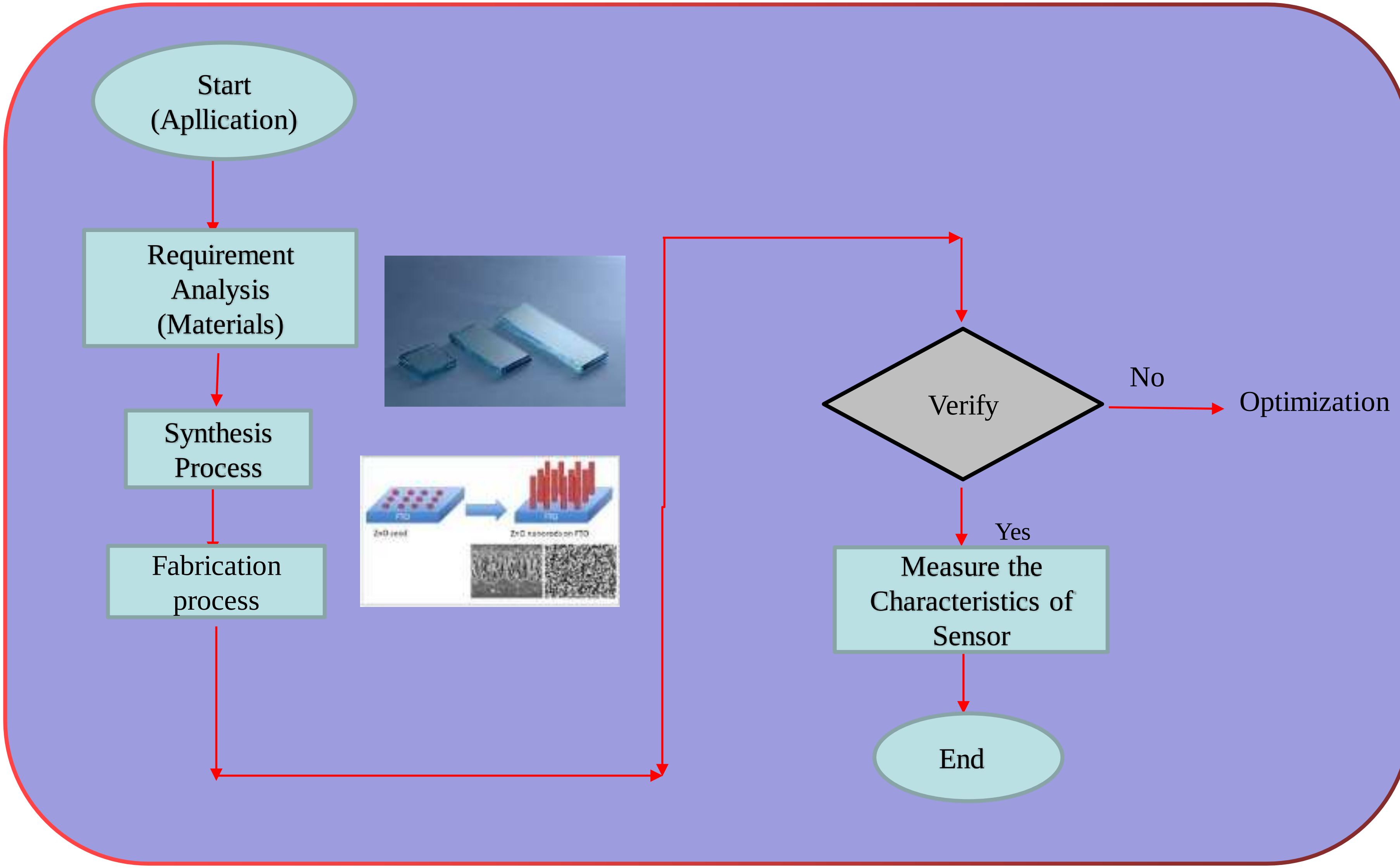
Literature Survey

S.NO	PUBLICATION DETAILS	MATERIAL/PARAMETER	INFERENCE
9	10.Senthilkumar mohan “Effective Heart Disease Prediction Using Hybrid Machine Learning Techniques” 2019	ML Algoriyhm/IoT 	• Effective assisting in making decisions and predictions from the large quantity of data produced by the healthcare industry • Improving accuracy in the prediction of cardiovascular disease
10	Xiu-Juan Xie “Synthesis of ZnO/NiO hollow spheres and their humidity sensing performance” 2021	ZnO/NiO/Response/recovery time 	• Oxygen vacancy and ion conduction • Improve humidity sensor performance:Synergistic effect between different components. • Smaller crystal size

Inferences from Literature Survey

- Surface area, porosity and grain sizes are contributing to the better sesnor response, i,e Sensitivity
- Morphology and Structure of a nanomaterials depend on reaction time, temperature and Concentration of reacting solutions
- By adapting the hydrothermal synthesis, i.e different reaction conditions and characterization, the shape of Zinc Oxide nanoparticles can be changed inorder to meet the requirements
- Change in resistance with respect to the variations in relative humidity can be measured to calculate the sensitivity of a humidity sensor. Respose and recovery time can also be moitored
- Band gap energy can be modulated through the concept of mixed metal oxide composites
- It is understood that, the sensitivity, linearity and long term stability can be improvised by introducing the material combination and also with suitable surface reactive agents
- As the particle size greatly affects the surface area of a nanomaterial, a humidity sensor's sensitivity may change

Workflow


Start
(Apllication)
Requirement
Analysis
(Materials)
Synthesis
Process
Fabrication
process
Verify
No
Optimization
Yes
Measure the
Characteristics of
Sensor
End

Time Line

	Months											
Activity	1	2	3	4	5	6	7	8	9	10	11	12
Literature Survey												
Fabrication of Sensor												
Experimental Analysis												
Testing												
Result												
Documentation												

Conclusion

The proposed research involves

- Fabrication and implementation of humidity sensor for respiratory monitoring application
- Herein, high sensitivity and linearity are the two important parameters considered
- Above said parameters can be achieved through proper selection of the composition of the material, reaction condition of synthesis
- Data moitoring and management may be carried out by ML algorithm and IoT

THANK YOU